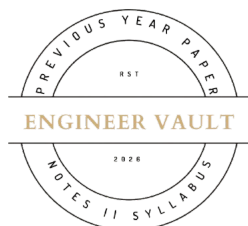


Total No. of Printed Pages: 2



SUBJECT CODE NO: - RNP-8005
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - III
Examination May/June-2026
Transformer and DC Machine

[Time: 2:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7.

PART A**Q1 Solve any 10 following questions****20**

- a) In DC series motor, speed is.....
 - a) Almost constant
 - b) Low at no-load
 - c) Very high at no-load
 - d) Zero at no-load
- b) A DC machine works on the principle of.....
 - a) Lenz's law
 - b) Faraday's law of electromagnetic induction
 - c) Coulomb's law
 - d) Gauss law
- c) The function of tap changer is to
 - a) Increase frequency
 - b) Control voltage
 - c) Reduce power factor
 - d) Increase current
- d) In an ideal transformer, input power is equal to.....
 - a) Zero
 - b) Output power
 - c) Copper loss
 - d) Core loss
- e) Efficiency of a transformer is maximum when
 - a) Core loss = Copper loss
 - b) Core loss > Copper loss
 - c) Copper loss = 0-meter
 - d) Total loss is minimum
- f) The core of a transformer is laminated to reduce
 - a) Hysteresis loss
 - b) Eddy current loss
 - c) Copper loss
 - d) Stray losses
- g) The transformer works on the principle of
 - a) Ohm's law
 - b) Mutual induction
 - c) Electrostatic inductions
 - d) Joule's law
- h) The starting torque of DC series motor is
 - a) Low
 - b) Medium
 - c) High
 - d) Zero
- i) Which DC generator has the poorest voltage regulation?

- a) Shunt generator
 - b) Series generator
 - c) Compound generator
 - d) Differential compound generator
- j) The main function of a yoke in a DC machine is.....
- a) Provide mechanical support
 - b) Complete the magnetic path
 - c) Protect the machine
 - d) All of the above
- k) The EMF generated in a DC generator is:
- a) AC
 - b) DC
 - c) Both AC and DC
 - d) None
- l) High starting current in DC motor is due to:
- a) High back EMF
 - b) Low armature resistance
 - c) High field flux
 - d) Mechanical load

PART B

Solve any four from the following

- Q.2** a) Explain Construction & Working principle of D.C generator 5
 b) Explain Construction & Working principle of 1Phase Transformer 5
- Q.3** a) Speed control of DC Shunt motor 5
 b) With the help of neat circuit diagram explain how O.C. & S.C. test is conducted on single-phase transformer. 5
- Q.4** a) Why is core of transformer laminated? Explain? 5
 b) What are the different types of losses in transformer & explain it? 5
- Q.5** a) Write a Short Note on Three-point starter 5
 b) Write a Short Note on Stepper Motor 5
- Q.6** a) A 4 pole DC Shunt generator with lap connected armature has field and armature resistance of $80\ \Omega$ and $0.1\ \Omega$ respectively. It supplies power to 50 lamps, rated to 100 volts, 60 watt each. Calculate total armature current and generated emf, by allowing a brush drop of 2V per brush. 5
 b) Explain: Power Flow Diagram of DC Machine 5
- Q.7** a) Explain: Losses in DC Machines. 5
 b) Explain: Armature Reaction in DC Machines. 5

